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In [2]: import numpy as np
import matplotlib.pyplot as plt

# Sample data (X: input, Y: output)
X = np.array([1, 2, 3, 4, 5])
Y = np.array([2, 4, 5, 4, 5])

# Calculate mean
x_mean = np.mean(X)
y_mean = np.mean(Y)

# Calculate slope (m) and intercept (b)
numerator = np.sum((X - x_mean) * (Y - y_mean))
denominator = np.sum((X - x_mean) ** 2)

m = numerator/denominator
b = y_mean - m * x_mean

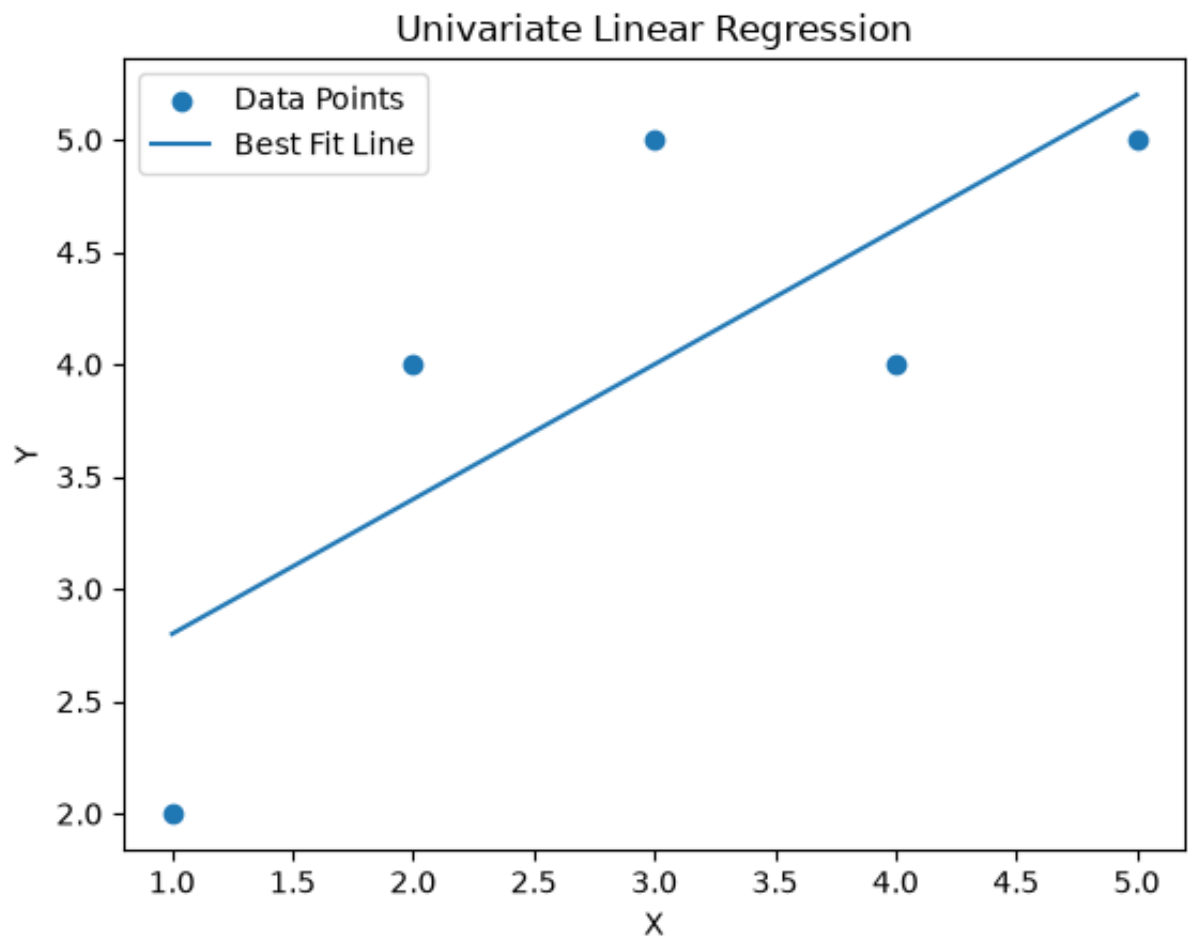
print("Slope (m):", m)
print("Intercept (b):", b)

# Predicted values
Y_pred = m * X + b
print("Value:", Y_pred)

x = input("Enter value: ")
yy = m * float(x) + b
print("Value:", yy)

# Plotting
plt.scatter(X, Y, label="Data Points")
plt.plot(X, Y_pred, label="Best Fit Line")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.title("Univariate Linear Regression")
plt.show()
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Slope (m): 0.6
Intercept (b): 2.2
Value: [2.8 3.4 4.  4.6 5.2]
Value: 5.8
```



In [ ]: